K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)
Semester: **August-December 2022**
In-Semester Examination

DE III & IV
CSM

Class: LY B. Tech
Branch: Computer
Full name of the course: Computer Simulation and Modeling
Course Code: UCEE704
Duration: 1hr.15 min

Semester : VII

Max. Marks: 30

Q. No	Questions	Marks	CO Mapped	BT Level
Q1	Consider a Refrigerator selling company. The inventory is maintained as by principle of order-upto-level (M) with review period of N days. The demand is distributed as shown in table. <table><tr><th>Demand</th><th>Probability</th></tr><tr><td>0</td><td>0.1</td></tr><tr><td>1</td><td>0.2</td></tr><tr><td>2</td><td>0.4</td></tr><tr><td>3</td><td>0.2</td></tr><tr><td>4</td><td>0.1</td></tr></table>The distribution of lead time is shown as following <table><tr><th>Lead Time</th><th>Probability</th></tr><tr><td>1</td><td>0.5</td></tr><tr><td>2</td><td>0.3</td></tr><tr><td>3</td><td>0.2</td></tr></table>Assume M = 10 and N = 4 days Let simulation start with inventory of 5 refrigerators and order of 5 refrigerators is expected to arrive in 3 days. Generate simulation table for period consisting of 2 review cycles. Find out how well the current (M, N) system is working with respect to shortage of refrigerators. R D for Demand: 7,5,3,2,9,0,3,4 R D for lead time: 8,5,2,4,1,9,2,1	10 marks	CO1,CO2	EV, CR
Q2	Define any 5 of the following with respect to Simulation and Modeling and give one example of each 1) Simulation 2) Model	10 marks	CO1	RE,UN,AP

<table>
<tr><td></td><td>3) System
4) System Environment
5) Entity
6) Attribute
7) Activity
8) State
9) Event
10) Endogenous
11) Exogenous</td><td></td><td></td><td></td></tr>
<tr><td>Q3</td><td>Students arrive at library with inter-arrival time distribution as follows,
<table><tr><td>Time between arrivals</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>probability</td><td>0.3</td><td>0.5</td><td>0.15</td><td>0.05</td></tr></table>
The time for transaction at counter is given by following distribution table,

RD: 55 ,25, 78, 89, 02, 67, 34, 21, 18, 93
<table><tr><td>Transaction Time</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>probability</td><td>0.2</td><td>0.4</td><td>0.25</td><td>0.15</td></tr></table>
RD: 34, 78 ,65, 11, 23, 28, 69, 89, 56, 28

If more than two students are in queue, arriving student goes away without joining the queue. Based on simulation of 10 students determine balking rate (leaving the queue).</td><td>10 marks</td><td>CO1, CO2</td><td>AN, EV, CR</td></tr>
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